

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is reported only as a surface-form reference; the main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level index Soft O2 improves over dense FlatIP, with hard hydration and shuffled-*sid* controls separating binding from score copying and membership noise. Gains are largest where a partial episode hit can pull in its answer sibling. Soft O2-C and gated Hybrid extend binding to BIRCH clusters under a Mix protocol with off-session gold; when query and gold already share a session, Soft O2 is preferred. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a complete/incomplete/miss failure taxonomy, and paired significance tests.

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1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology; neighbouring matters are topically adjacent yet legally distinct; and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017). Recent legal large language models (LLMs) improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR) mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023). The gap addressed here is *internal consultation memory*: retrieving prior client-specific advice from a growing same-domain archive under lexical interference.

BIMS-LEGAL (Brain-Inspired Memory System for legal consultation archives) separates a fast episodic pathway—timestamped turn embeddings with speaker role and session identifier *sid*—from a slow semantic pathway based on Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) with deferred consolidation Zhang et al. (1996). Complementary Learning Systems (CLS) theory McClelland et al. (1995); Kumaran et al. (2016) is used only as an architectural analogy for that dual-pathway split. Under same-domain interference, dense turn retrieval in a vector database often returns a question turn while its answer sibling remains outside top- k —a system-level episodic memory failure we term *episode incompleteness*, distinct from a full session miss and exacerbated by IVFPQ score collapse and context fragmentation at moderate archive sizes.

34 The *algorithmic contributions* are soft binding operators that complete
 35 episodes without hard score copy:

- 36 • **Soft O2 (session soft binding).** When a turn is retrieved with score
 37 s , siblings sharing *sid* inherit $\max(s_t, \beta \cdot s)$, surfacing answer turns
 38 under question-hit / answer-miss while preserving rank competition
 39 among candidates.
- 40 • **Soft O2-C and gated Hybrid (cluster soft binding).** The same
 41 soft-inheritance rule applies within BIRCH clusters when gold evidence
 42 is not co-session with the query; Hybrid triggers cluster binding only
 43 on direct dense hits to avoid displacing same-session siblings.

44 Two **implementation operators** support large shared-store experiments
 45 but are not standalone algorithmic contributions:

- 46 • **O1 (exact FlatIP indexing).** At hundreds-to-thousands of turns,
 47 IVFPQ product quantization is under-trained and collapses Answer
 48 Hit; FlatIP restores faithful turn scores so Soft O2 inherits uncorrupted
 49 sibling signals.
- 50 • **O3 (bulk-load consolidation).** Deferred BIRCH and disk writes
 51 make archive construction tractable without changing retrieval seman-
 52 tics.

53 We ask four research questions:

- 54 • **RQ1.** Under same-domain interference, how prevalent is episode in-
 55 completeness (question hit / answer miss) relative to full session miss?
- 56 • **RQ2.** Do Soft O2 and Soft O2-C address episode incompleteness under
 57 measured index distortion, and what incremental gains does Soft O2
 58 provide over FlatIP and hard parent hydration?
- 59 • **RQ3.** On the *same* LegalEp /LegalMem stores, when is the slow
 60 (BIRCH) pathway effective? Does Soft O2-C help under standard same-
 61 session gold, and under a fair Mix of same-/cross-session gold?
- 62 • **RQ4.** Across CAIL multi-turn and LegalEp paraphrase /advice-recall
 63 channels, how large and session-tied are Soft O2 gains, and when do
 64 Soft O2-C /Hybrid help beyond Soft O2?

65 *Contributions..*

- 66 1. A dual-store memory architecture for Chinese legal consultation archives,
67 separating fast episodic turn recovery from slower BIRCH semantic in-
68 tegration under same-domain interference.
- 69 2. Soft O2, a turn-level session soft-binding operator that recovers missing
70 answer siblings without hard parent hydration or session-max score
71 copy.
- 72 3. A three-candidate score-competition account of Soft O2 using the gap
73 ratio $\rho = s_d/s_h$, with default $\beta=0.98$ chosen by validation sweep.
- 74 4. Turn-level lexical and fusion controls (BM25-turn, dense+BM25 RRF,
75 cross-encoder) under the same evidence unit as Soft O2, with joint
76 episode indexing treated as a coarser alternative design rather than a
77 turn-binding baseline.
- 78 5. Soft O2-C and gated Hybrid on Mix cross-session reconstructions, graded
79 nDCG with fixed gold-session IDCG, a failure taxonomy, and paired
80 significance tests.
- 81 6. O1 exact FlatIP and O3 bulk-load consolidation as implementation
82 operators for the shared-store ablations.

83 Section 2 reviews related work. Section 3 presents BIMS-LEGAL opera-
84 tors and the Soft O2 attenuation analysis. Section 4 details corpora, metrics,
85 and protocols. Section 5 reports results. Section 6 discusses implications and
86 limitations. Section 7 concludes.

87 2. Related Work

88 2.1. Legal IR and long-horizon dialogue memory

89 Legal QA and retrieval systems typically ground generation in statutes,
90 cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-
91 Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware
92 similar-case ranking further show how interaction context and structured le-
93 gal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024).
94 Those pipelines optimize access to external authority. Consultation memory
95 retrieval targets a different evidence source: *prior consultation episodes* inside
96 an agent’s own archive. Confusing two clients’ histories can produce fluent
97 yet professionally inappropriate responses even when statute retrieval is cor-
98 rect. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM

improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question-answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. Long-MemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and BIRCH centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure Zhang et al. (1996). Under dense same-domain interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing. An alternative way to keep episode context is to concatenate full question-answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2

denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH path-way.

In short, the dual-store split is the system design, and Soft O2 /Soft O2-C are the operators evaluated on Chinese legal consultation archives.

3. Method: BIMS-LEGAL

3.1. Architecture overview

Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complementary episodic and semantic stores over the same consultation archive.

Definition 1 (Episodic and semantic elements). *An episodic turn (episodic element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a timestamp, c_i conversational context (including speaker role), ϕ_i surface features, and sid_i the consultation session identifier. We reserve session turns for the weaker notion of turns that share only sid under a grouping key, without requiring role, timestamp, or other episodic metadata to participate in scoring. Soft O2 binds over session siblings $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$, but the indexed objects themselves remain episodic turns. A semantic element is $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation statistics τ_j . Write $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ for BIRCH-cluster siblings.*

Definition 2 (Dual-store retrieval map). *Given query q with embedding $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$ under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Section 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Section 3.5). Gated Hybrid composes session binding then cross-session cluster binding (Algorithm 2). The returned list is the top- k of the bound set under the rewritten scores.*

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

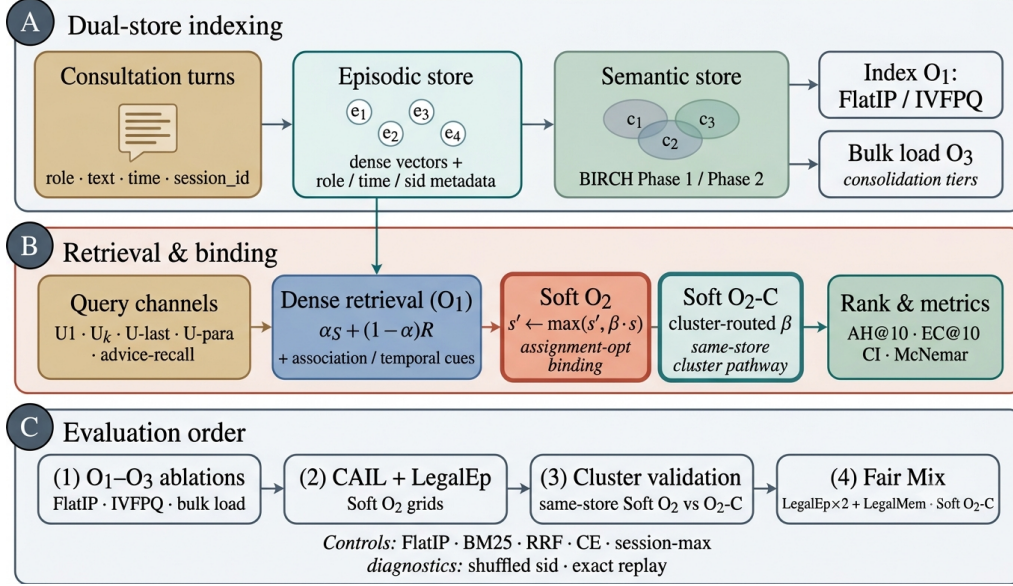


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

191 with Δ_t the elapsed time between query time and the turn timestamp. Unless
 192 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
 193 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
 194 and do not replace Eq. (1).

195 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 196 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 197 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 198 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 199 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 200 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 201 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 202 algorithm.

203 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 204 insert update ($e_i \mapsto \text{BIRCH}(e_i)$; Flush) by a deferred operator $\text{Finalize} =$
 205 $\text{BIRCH}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic writes. O3 does
 206 not change retrieval semantics; it makes $M \geq 400$ shared-store experiments

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

runnable (≈ 11 min wall-clock for $M=400$; amortized ≈ 0.825 s/turn over
 ~ 800 turns). Quality–latency curves to $M=6400$ appear in Section 5.6.

3.3. Soft O2: session soft binding

Motivation.. Under same-domain interference, a retrieved question turn can score highly while its answer sibling remains outside top- k —episode incompleteness compounded by IVFPQ score collapse and context fragmentation. Soft O2 completes episodes without hard score copy.

Definition 3 (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is the special case $\beta=1$ with forced insertion of all siblings; session-max replaces each member score by $\max_{t \in sid} s_t$.

Algorithm 1 states Soft O2 as sibling expansion followed by a completeness pass and top- k truncation, matching the public implementation.

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). Default β is chosen by the AH sweep in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

221 3.4. Soft O2 attenuation and score competition

222 The attenuation β controls how strongly a retrieved turn promotes its
 223 session siblings. Availability of the injected sibling rises with β , while fidelity
 224 of the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-ratio
 225 distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy:
 226 median ρ lies well below one, and a high- β operating point near 0.98 covers
 227 most measured distractors while remaining strictly below hard hydration
 228 ($\beta=1$). We therefore characterize the availability-ranking trade-off with a
 229 three-candidate score-competition model and choose β by validation sweep
 230 rather than by a closed-form optimum. A zero-margin hinge objective on
 231 $\beta \in (0, 1]$ is uninformative for this choice: its rank term vanishes for all
 232 $\beta < 1$.

233 **Definition 4** (Three-candidate score-competition model). *For a query whose*
 234 *dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be*
 235 *the gold answer sibling and let d be the strongest non-session distractor in a*
 236 *wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

237 *and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured*
 238 *distractor, conditions based on ρ are conservative sufficient conditions for*
 239 *beating that distractor, not exact top- k boundary conditions for every com-*
 240 *petitor.*

241 The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$,
 242 where $s'_a = \beta s_h$.

243 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*
 244 *the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured*
 245 *distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the*

246 *injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$,*
 247 *any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$*
 248 *already, rank preservation depends on the observed sibling score rather than*
 249 *on β alone.*

250 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 251 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 252 $s_h > 0$. $\square$

253 *Soft-rank visualization and sweep..* On a validation grid we also plot the soft
 254 pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

255 with logistic σ , temperature τ , and rank weight γ . The curve is a visualization
 256 of the availability–ranking trade-off under the measured ρ pool; it is not used
 257 as a fitted optimum. Default $\beta=0.98$ is taken from the AH sweep in Table A.4,
 258 where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band on CAIL follow-up and
 259 LegalEp paraphrase as well as exact grids. Advice-recall uses the same default
 260 without a separate gap-ratio fit.

261 Fair controls include hard parent hydration, session-max aggregation, and
 262 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 263 coincide under full-sibling expansion, so tables report a single hard-expansion
 264 baseline. Concatenating a full question–answer episode into one retrieval unit
 265 (BM25-joint or dense joint-episode documents) changes the evidence unit
 266 from turns to documents; we treat that design as an alternative indexing
 267 choice rather than a Soft O2 turn-binding baseline (Section 4).

268 3.5. Semantic consolidation and Soft O2-C

269 Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang
 270 et al. (1996).

271 **Definition 5** (BIRCH dual-phase consolidation). *Let $\{\mathbf{c}_j\}$ be current cen-*
 272 *troids and θ_H, θ_M the assign /merge thresholds. **Phase 1 (assign).** For*
 273 *a new embedding $\mathbf{v}$, set $j^* = \arg \max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^*}) \geq \theta_H$ then*
 274 *$C_{j^*} \leftarrow C_{j^*} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^*}$, else spawn a new cluster. **Phase 2***
 275 *(merge). Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-*
 276 *duplicates above a redundancy threshold. Under O3, both phases run inside*
 277 *Finalize. Each turn t receives a cluster id cid_t , inducing $\text{Clus}(t)$ in Defini-*
 278 *tion 1.*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

279 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
280 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
281 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
282 *thematic clusters cannot flood top- k .*

283 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
284 relative to sessions already covered by dense /Soft O2 hits, and triggers
285 cluster binding only on direct dense hits. This prevents thematic confusers
286 from displacing same-session siblings already recovered by Soft O2. Soft O2
287 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
288 operator when gold evidence lies in another session of the same thematic
289 cluster.

290 4. Corpora and Evaluation Protocol

291 Figure 2 sketches the shared-corpus design used throughout the legal eval-
292 uation. Gold needles and same-domain distractors occupy one store. Each
293 query probes that shared archive under fixed interference conditions. Pri-
294 vately rebuilt haystacks are avoided so that system comparisons isolate the
295 retrieval operator.

296 4.1. Datasets and query channels

297 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
298 Gold needles are authentic preliminary and final dialogues, with conversation
299 turns completed by label turns where applicable. Query channels cover the

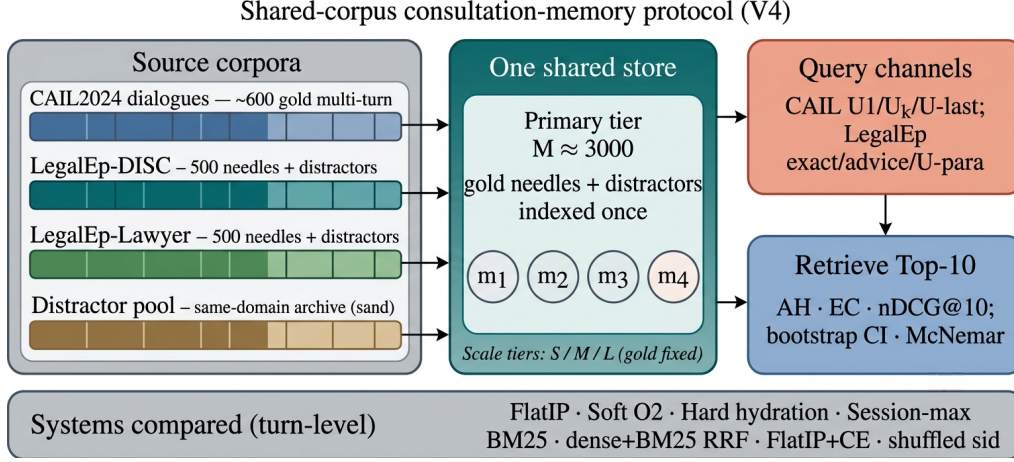


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

300 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 301 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 302 assistant turns in the target session.

303 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 304 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 305 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 306 applies length bounds and legal-content filters, excludes exam prompts, re-
 307 moves near-duplicates, separates needles from distractors under a fixed seed,
 308 and retains an audit subsample. After filtering, each LegalEp corpus retains
 309 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 310 tier).

311 *Query-channel policy.* Exact replay uses the stored user question and is
 312 reported only as a *diagnostic* upper bound on surface-form recovery. Primary
 313 claims use paraphrase, $U_k\text{-followup}$ / $U\text{-last}$, and advice-recall queries whose
 314 surface form is not identical to the stored question text; paraphrase and
 315 advice-recall templates are generated under a fixed seed and audited on a
 316 subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store
 317 for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

318 Archive size is varied with gold needles held fixed. Unless stated otherwise,
 319 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,

320 $S=300$.

321 4.2. Metrics and baselines

322 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
323 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is
324 binary: whether a gold answer turn appears in the top- k . EC is the mean
325 coverage of gold turns within the target session. Session Hit@ k is reported
326 only for diagnosis. For nDCG we use fixed graded relevance over the full
327 gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
328 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
329 gold-relevant turns in the target session and truncated at k , so every system
330 on a query shares the same IDCG. Answer Hit and Episode Completeness
331 are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session
332 IDCG so that systems remain comparable on each query. Hard expansion
333 can raise nDCG when copied siblings occupy early ranks, so AH and EC are
334 read together with nDCG.

335 The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL
336 and LegalEp transfer channels; for that family we report bootstrap 95% CIs,
337 McNemar mid- p , and Holm-adjusted p -values (Table A.2). Hard-hydration,
338 CE, and Soft O2-C /Hybrid Mix contrasts are secondary.

339 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
340 soft sibling score inheritance on the episodic pathway; Soft O2-C applies
341 the same rule within BIRCH clusters. Hard parent hydration expands a hit
342 to all siblings by copying the trigger score. Lexical controls include BM25
343 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same
344 turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve
345 pre-concatenated episode documents and therefore answer a coarser retrieval
346 problem; they are discussed as alternative designs in Section 6.3 rather than
347 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the
348 top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3. Shuffled *sid*,
349 a β sweep, and IVFPQ provide membership, attenuation, and quantization
350 controls.

351 4.3. Fair Mix protocol for Soft O2-C

352 On fully same-session LegalMem grids Soft O2 already recovers gold an-
353 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
354 ing it to thematic confusers. To study off-session recovery we rebuild the *same*
355 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes

are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store; we report overall AH and stratum AH on `cross_session` vs `same_session` queries.

Cluster co-membership and purity under Mix. After BIRCH consolidation, each cross-session S_a/S_b pair is assigned a shared cluster identifier. This construction uses split metadata available only when building the evaluation store; it does not inject labels at query time. The intent is to test Soft O2-C when thematic co-membership is present. Cluster purity still matters: a shared *cid* that also contains many thematically adjacent distractors narrows Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to direct dense hits and caps the per-cluster budget B_c . If recovery still fails under guaranteed S_a/S_b co-membership, the failure is attributable to binding rather than accidental cluster separation. Natural same-cluster rates without this construction are lower (Section 6.3). Soft O2’s session membership rule is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon memory context only Wu et al. (2024); Maharana et al. (2024).

5. Experiments

Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe` Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with $k=10$ and seed 42 on the main $M \approx 3000$ stores. We keep the encoder fixed so Soft O2 vs FlatIP contrasts isolate binding rather than representation changes; sensitivity to alternative Chinese or multilingual encoders (e.g., `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a smaller shared store ($M=400$) is reported later. Primary contrasts appear in Figures 3–5; full numeric grids are in the appendix. Each main table maps to a released JSON artifact in the accompanying repository.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone

389 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 390 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 391 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 392 over baseline) and Episode Completeness to 0.888 / 0.840. O3 is enabled for
 393 all configurations in this ladder as a construction prerequisite; it does not
 394 alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

395 5.2. RQ1 failure taxonomy

396 We classify each query into *complete* (answer evidence in top- k), *incom-*
 397 *plete* (target session partially retrieved but answer missing), or *session miss*
 398 (no target-session turn in top- k). Table 4 contrasts FlatIP, Soft O2, hard
 399 hydration, and shuffled *sid* on primary non-exact channels. Soft O2 converts
 400 incompleteness into complete retrieval while session-miss stays nearly fixed;
 401 shuffled *sid* restores high incompleteness, so the gain is session-tied rather
 402 than generic score inflation. Hard hydration can also drive incompleteness
 403 near zero by copying trigger scores, but it is not Soft O2’s binding rule and
 404 can over-promote siblings (Section 5.4).

405 5.3. Turn-level Soft O2 controls

406 Primary Soft O2 claims use a shared turn-level index. Table 5 reports An-
 407 swer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary
 408 Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel;
 409 shuffled *sid* collapses toward or below FlatIP; hard hydration is competitive
 410 on CAIL AH but Soft O2 remains stronger on LegalEp paraphrase /advice
 411 and retains higher EC in the full grids (Tables A.5–A.7). BM25-turn is a

Table 4: RQ1 failure taxonomy ($k=10$). Boldface: best complete / lowest incomplete within each corpus-channel block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Soft O2 cuts incompleteness without raising session miss; shuffled *sid* undoes the gain.

Corpus / channel	System	Complete	Incomplete	Session miss
DISC / u-param	FlatIP	70.8%	26.8%	1.6%
	Soft O2	89.1%	8.7%	1.8%
	Hard hydr.	97.2%	0.0%	2.8%
	Shuffled O2	66.4%	31.4%	2.2%
DISC / advice	FlatIP	48.0%	17.0%	35.0%
	Soft O2	60.8%	3.8%	35.4%
	Hard hydr.	65.0%	0.0%	35.0%
	Shuffled O2	45.6%	19.2%	35.2%
Lawyer / u-param	FlatIP	71.5%	25.3%	0.8%
	Soft O2	94.0%	3.8%	1.6%
	Hard hydr.	98.0%	0.0%	2.0%
	Shuffled O2	72.9%	25.5%	1.6%
CAIL / Uk	FlatIP	45.8%	54.0%	0.2%
	Soft O2	92.8%	7.0%	0.2%
	Hard hydr.	98.3%	1.5%	0.2%
	Shuffled O2	34.7%	65.2%	0.2%

412 same-granularity lexical control (Table A.8): Soft O2 exceeds it on CAIL
 413 Uk /U-last (0.698/0.762 vs 0.482/0.413). Joint episode / document-level in-
 414 dexes change evidence granularity and are not Soft O2 protocol competitors.

415 5.4. Soft O2 on session (CAIL and LegalEp)

416 Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn
 417 channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP
 418 on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs
 419 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200),
 420 tying the improvement to correct episode membership (RQ1–RQ2). Hard
 421 expansion can raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788)
 422 while lowering EC (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves
 423 as a rank-preserving binder: siblings enter under attenuated scores, and
 424 episode coverage remains higher than under hard score copy. Table 6 shows
 425 the Soft O2>FlatIP ordering under graded nDCG with gold-session IDCG,

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is the session-binding claim; shuffled *sid* tests membership dependence; hard hydration is score-copy expansion.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.570	0.788	0.802	0.452
CAIL / Uk	0.270	0.698	0.720	0.230
CAIL / U-last	0.245	0.762	0.790	0.200
LegalEp-DISC / u-para	0.547	0.648	0.592	0.469
LegalEp-DISC / advice	0.342	0.428	0.402	0.310
LegalEp-Lawyer / u-para	0.568	0.759	0.596	0.592
LegalEp-Lawyer / advice	0.332	0.472	0.394	0.348

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

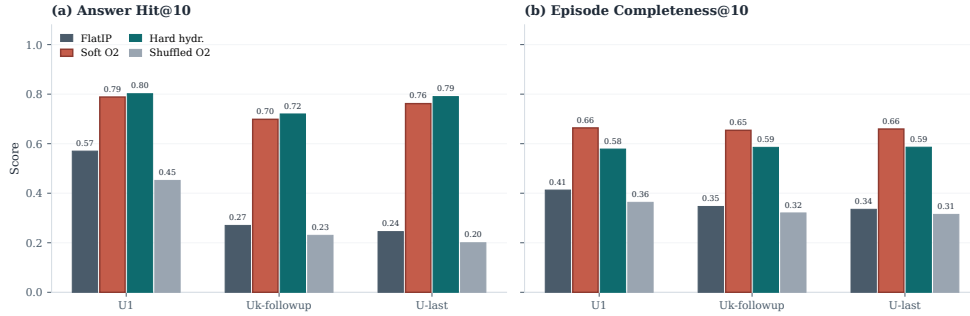


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where both turns are already recoverable for many queries—consistent with treating exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752 from FlatIP 0.692 (McNemar mid- $p=0.024^*$). Paraphrase yields the clearest LegalEp gains: Soft O2 reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs 0.568 on Lawyer, exceeding hard expansion on both corpora (mid- $p<0.001^{**}$ on both corpora). Advice-recall queries omit the stored question surface form: Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from 0.332 on Lawyer, again above hard expansion (mid- $p<0.001^{**}$), while shuffled

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	0.793	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	0.791	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.759	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.524	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.844	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.582	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

438 *sid* falls toward or below FlatIP. Graded nDCG mirrors the Soft O2>FlatIP
 439 ordering on paraphrase and advice-recall (Table 6). Detailed grids are in
 440 Tables A.6–A.7.

441 Cross-encoder reranking is a strong neural control at the same turn-
 442 level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with
 443 **bge-reranker-v2-m3**. Table 7 shows that FlatIP+CE is competitive—and
 444 on LegalEp-DISC exact, strongest—when surface-form overlap already places
 445 both question and answer near the top of the dense pool (DISC exact AH
 446 0.722 for FlatIP+CE vs 0.638 for Soft O2). On channels with semantic shift
 447 or multi-turn drift, Soft O2 pulls ahead: CAIL Uk 0.698 vs FlatIP+CE 0.375,
 448 CAIL U-last 0.762 vs 0.342, and LegalEp-Lawyer exact 0.752 vs 0.708. A
 449 structural limit of the reranker explains the split: CE can refine ranking
 450 among candidates already retrieved by FlatIP, but it cannot promote an
 451 answer sibling that never entered the top-30 pool. Soft O2 recovers such
 452 siblings by propagating attenuated scores through session links, which helps
 453 precisely when the stored advice turn has weak direct lexical overlap with

LegalEp ($M=3000$): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

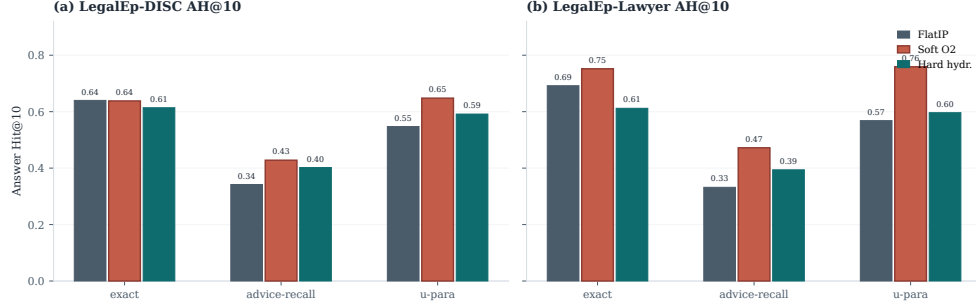


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

454 the query (paraphrase and advice-recall; cf. Tables 5 and 6).

455 5.5. Soft O2-C and gated Hybrid under Mix

456 When gold answers already share *sid* with the query, Soft O2 is sufficient
457 and cluster binding is unnecessary. Table 8 asks the off-session question
458 under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and
459 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated
460 Hybrid sharing unrestricted dense retrieval.

461 On the cross-session stratum Soft O2-C improves over Soft O2 on LegalEp-
462 Lawyer (0.491 vs 0.446) and LegalMem-MT (0.537 vs 0.503): cluster binding
463 can surface split-episode gold that session membership alone misses. Gated
464 Hybrid converts that mechanism into an overall gain on LegalEp-Lawyer
465 (0.534 vs Soft O2 0.496; mid- $p=0.010^*$) and a directional gain on LegalMem-
466 MT (0.646 vs 0.618), while preserving most same-session hits (0.560 / 0.867).
467 On LegalEp-DISC the Hybrid edge is negligible (0.486 vs 0.482). Ungated

468 Soft O2-C can raise cross-session hits yet displace intact same-session evi-
469 dence, which is why Hybrid triggers cluster binding only on direct dense
470 hits. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Ap-
471 pendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p < 0.05$, ** $p < 0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

472 McNemar mid- p on paired Answer Hit; * $p < 0.05$, ** $p < 0.01$. Cluster co-membership
473 of Sa/Sb pairs follows the Mix protocol in Section 4.3.

474 5.6. Scale behaviour

475 Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-
476 DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2
477 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH
478 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP
479 as an engineering prerequisite for Soft O2 at these scales and avoid claiming
480 deployment readiness beyond the measured curve.

481 *End-to-end QA audit (summary)*.. A dual-protocol QA audit ($N=270$ per
482 corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged
483 answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs
484 Soft O2 generation experiment. Full protocol and stratified results appear in

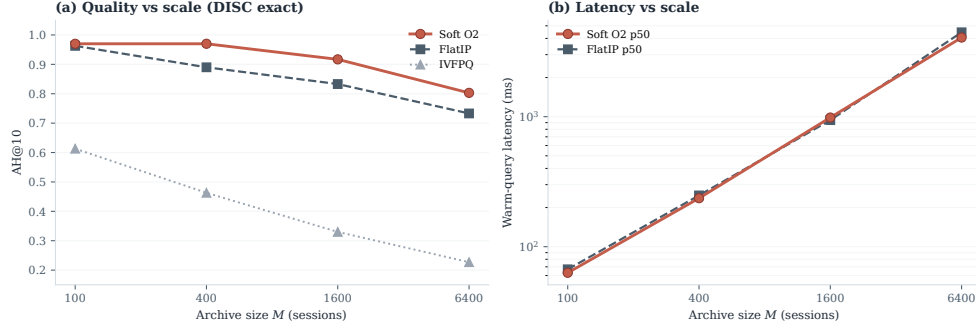


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

6. Discussion

6.1. Findings and scope

RQ1–RQ2 show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. After Holm adjustment on the Soft O2 vs FlatIP family, the non-exact CAIL and LegalEp gains remain significant, while LegalEp exact replay is mixed (Table A.2). LegalEp paraphrase and advice-recall, rather than exact replay, support the main transfer result. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid recovers additional cross-session gold on LegalEp-Lawyer with a directional lift on LegalMem-MT (Table 8); Soft O2 remains preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robert-

son and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder `legal_qa_n270`). The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score ≥ 0.7 . Table 9 reports pooled EAR and the mean judge score on a stratified subsample ($n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Stratified judge” in the table. In the complete-episode stratum—both question and answer turns in top- k —judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval failures (gold episode absent from top- k) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

542 *Dependency on session segmentation and cluster quality..* Soft O2 relies on
543 correct session identifiers (*sid*). In uncured client logs, boundary noise—
544 mid-session topic shifts, multi-intent interruptions, or automated segmenta-
545 tion drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated
546 inheritance may propagate false context; if a true advice turn is split into
547 another session, Soft O2 cannot bind it. Soft O2-C likewise depends on
548 BIRCH cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared
549 cluster id so binding can be tested when thematic co-membership is present
550 (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink
551 the margin over dense retrieval. Deployments on raw consultation streams
552 therefore need reliable upstream session segmentation, and Soft O2-C should
553 be treated as gated rather than always-on.

554 *Joint-episode indexing..* Concatenating full question–answer pairs into
555 episode documents prevents incompleteness by changing the evidence unit
556 from turns to merged documents (Section 2). That design is outside the
557 Soft O2 turn-binding protocol; a matched comparison under online updates,
558 latency, and fine-grained advice isolation is left to future work.

559 *Embedding model and other scope limits..* All main grids use
560 `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as
561 `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ
562 distribution, and might shrink or widen Soft O2’s relative gain wherever
563 the bi-encoder already places answer turns inside top- k . We did not re-run
564 the full Soft O2 grids under alternative encoders; the binding operator
565 itself is encoder-agnostic once fused scores are available. The ρ statistics
566 use an exact-channel query proxy and the strongest measured distractor;
567 paraphrase and advice-recall gap laws may shift the feasible band slightly,
568 though the AH sweep keeps a stable high- β region on those channels. Main
569 $M \approx 3000$ tables use a fixed seed; five-seed variance is reported only for the

570 smaller $M=400$ store. LLM paraphrases need human audit. Statutory
 571 correctness, citation validity, and professional liability remain unevaluated.
 572 Because many channels are tested, Holm-adjusted p -values accompany
 573 the Soft O2 vs FlatIP family (Table A.2); Hybrid Mix gains outside
 574 LegalEp-Lawyer are treated as directional.

575 *Latency and deployment scope.* Scale curves cover $M \in$
 576 $\{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At
 577 $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms
 578 for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL
 579 should therefore be positioned as **high-precision offline or quasi-**
 580 **real-time consultation memory recovery**—e.g., pre-meeting case
 581 review, audit replay, or batch RAG over a firm’s advice archive—not as
 582 a sub-millisecond search engine. Future work will combine approximate
 583 nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C
 584 binding layered on top of faithful candidate retrieval, preserving episode
 585 completion while reducing query latency. Behavior at roughly 10^5 sessions is
 586 not measured, and we do not claim asymptotic scalability from O3 wall-clock
 587 alone.

588 7. Conclusion

589 BIMS-LEGAL recovers prior advice from Chinese legal consultation
 590 archives under same-domain interference with a dual store: an episodic turn
 591 index and a BIRCH semantic store. Soft O2 is the main algorithmic con-
 592 tribution: attenuated session binding completes episodes without hard score
 593 copy. Soft O2-C and gated Hybrid extend the same idea to thematic clus-
 594 ters when Mix reconstructions place some gold off-session. Attenuation β
 595 is chosen by validation sweep under a three-candidate score-competition ac-
 596 count; O1 FlatIP and O3 bulk-load are the implementation operators that
 597 make the shared-store ablations feasible. On CAIL multi-turn and LegalEp
 598 paraphrase /advice-recall channels, Soft O2 improves Answer Hit over FlatIP,
 599 keeps higher episode completeness than hard expansion on CAIL, and raises
 600 graded nDCG while cutting incompleteness; shuffled-*sid* controls collapse
 601 the gains. Under Mix, gated Hybrid recovers additional cross-session gold on
 602 LegalEp-Lawyer while Soft O2 remains preferable on intact same-session gold.
 603 An end-to-end Soft O2 QA audit associates complete episode retrieval with
 604 judged answerability and low wrong-session contamination, without claim-
 605 ing paired generation superiority over FlatIP. Turn-level BM25, RRF, and

606 CE place Soft O2 among standard IR alternatives; joint-episode indexing
607 remains a coarser design outside this protocol.

608 **Declarations**

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610 Computational resources were provided by the Data Law Laboratory,
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612 *Conflict of interest*

613 The authors declare no competing interests.

614 *Ethics approval*

615 This study uses publicly released research corpora and excludes confiden-
616 tial client data. The system is a research prototype for consultation-memory
617 retrieval.

618 *Data availability*

619 Code, processed evaluation corpora, and primary result JSON files are
620 available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory> (archived release to accompany camera-ready). Upstream
622 corpora remain under their original licenses: CAIL2024 consultation tracks;
623 Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_d
624 ata. Processed LegalEp /LegalMem artifacts are derived research releases;
625 redistributors should consult each upstream licence before reuse.

626 *Code availability*

627 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,
628 Soft O2-C), corpus builders, Mix constructors, and table/figure scripts ship
629 in the same repository, with one JSON artifact per main table.

630 *Author contributions*

631 L.X. (University of Winnipeg) designed BIMS-LEGAL and the
632 Soft O2 /Soft O2-C evaluation, implemented the system, ran the experi-
633 ments, and wrote the manuscript. L.H. (corresponding author; CUPL Data
634 Law Lab and Institute for Data Law) provided methodology guidance, revi-
635 sion, and supervision.

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754 **Appendix A. Numeric grids and controls**

755 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 756 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.554	[0.530,0.610]
	Soft O2	0.788	0.664	0.776	[0.757,0.820]
	Hard hydr.	0.802	0.578	0.864	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.411	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.445	[0.237,0.307]
	Soft O2	0.698	0.654	0.793	[0.660,0.735]
	Hard hydr.	0.720	0.585	0.847	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.369	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.420	[0.212,0.280]
	Soft O2	0.762	0.659	0.791	[0.725,0.797]
	Hard hydr.	0.790	0.586	0.854	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.348	[0.172,0.232]

757

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	0.730
	Soft O2	0.638	0.780	0.813
	Hard hydr.	0.614	0.598	0.886
	Shuffled O2	0.448	0.645	0.666
advice-recall	FlatIP	0.342	0.438	0.466
	Soft O2	0.428	0.503	0.524
	Hard hydr.	0.402	0.405	0.572
	Shuffled O2	0.310	0.421	0.436
u-param	FlatIP	0.547	0.681	0.681
	Soft O2	0.648	0.735	0.759
	Hard hydr.	0.592	0.571	0.827
	Shuffled O2	0.469	0.610	0.627

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	0.753
	Soft O2	0.752	0.848	0.869
	Hard hydr.	0.612	0.615	0.897
	Shuffled O2	0.570	0.747	0.705
advice-recall	FlatIP	0.332	0.464	0.483
	Soft O2	0.472	0.540	0.582
	Hard hydr.	0.394	0.403	0.601
	Shuffled O2	0.348	0.467	0.454
u-param	FlatIP	0.568	0.718	0.744
	Soft O2	0.759	0.819	0.844
	Hard hydr.	0.596	0.594	0.864
	Shuffled O2	0.592	0.712	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460